

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$= \{\chi^{\mu\nu}\} + \begin{pmatrix} ((p^2 - q^2)a + 2pqb) & (pa + qb) & (pb - qa) & ((p^2 - q^2)a + 2pqb) \\ (pa + qb) & 0 & 0 & (pa + qb) \\ (pb - qa) & 0 & 0 & (pb - qa) \\ ((p^2 - q^2)a + 2pqb) & (pa + qb) & (pb - qa) & ((p^2 - q^2)a + 2pqb) \end{pmatrix}.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\sigma^* = \begin{cases} \ell + \alpha & 0 < \gamma < \frac{\ell + \alpha}{d} + \frac{1}{p}, \\ \frac{d}{d-1} \left(\ell + \alpha + \frac{1}{p} - \gamma \right) & \frac{\ell + \alpha}{d} + \frac{1}{p} \leq \gamma < \ell + \alpha + \frac{1}{p}, \end{cases}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$z_t = \begin{pmatrix} x_t \\ y_t \end{pmatrix} = U_t(T, D) \begin{pmatrix} x'_1 \\ y'_1 \end{pmatrix} - D U_{t-1}(T, D) \begin{pmatrix} x'_0 \\ y'_0 \end{pmatrix} + \begin{pmatrix} x^* \\ y^* \end{pmatrix}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \frac{df}{dt} &= \frac{\partial f}{\partial t} + \{f, H_T\} \\ &= \frac{\partial f}{\partial t} + \{f, H_T\}_D + \{f, \chi_s\} (\Delta^{-1})^{ss'} \{\chi_{s'}, H_T\} \approx \frac{\partial f}{\partial t} + \{f, H_T\}_D. \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} a^{A*} &= a^A & b^{A*} &= b^A & \omega_{AB}^* &= \omega_{AB} & \lambda^* &= \lambda & T^\dagger &= -T \\ \bar{\rho}_i &= \rho^{i\dagger} \gamma^0 & \bar{\varepsilon}_i &= \varepsilon^{i\dagger} \tilde{\gamma}^0 \end{aligned}$$